

John Graham Reynolds

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TECHNICAL EXPERTISE

Knowledge System : RAG, GraphRAG, Hybrid Vector + Knowledge Graph Retrieval, Neo4j, Semantic Search, LLM Agents, Agentic Information Retrieval

AI/ML Expertise : Transformers, NLP, Language Modeling, Reinforcement Learning, Generative AI, Fine-tuning, CNNs, Computer Vision

Coding & OS : Python, Mathematica, C/C++, SQL, Bash/Zsh, LaTeX, Git, Docker, Linux/macOS/Windows

AI/ML Frameworks : PyTorch, Hugging Face, LangChain, Gymnasium, Ollama, vLLM, Qdrant, DeepEval, Cirq, Streamlit, Playwright, FastAPI

Cloud & Platform : Azure, Databricks, Terraform, Delta Lake, Apache Spark, GitHub/GitLab, HPC Systems, Nvidia Blackwell/Ada Lovelace/Ampere/Turing, Google TPUs

Quantum & Physics : Tensor Networks, Quantum Computing, Quantum Information Theory, General Relativity, Quantum Field Theory, String Theory

EDUCATION

University of Texas at Austin

Austin, TX

Master of Science in Artificial Intelligence - Department of Computer Science

Aug 2025 – May 2027, expected

- **Relevant Coursework:** Advanced Deep Learning, Deep Learning, Natural Language Processing, Reinforcement Learning
- **Research Focus:** Agentic Systems for Scientific Research: Mathematics, Theoretical Physics, Auto-improving AI

Johns Hopkins University

Baltimore, MD

Bachelor of Science in Physics, Mathematics

Aug 2016 – May 2020

- **Advanced Physics Research:** Quantum Mechanics, Quantum Field Theory, General Relativity, String Theory
- **Advanced Mathematics Research:** Differential Geometry, Abstract Algebra, Complex Analysis, Tensor Calculus

PROFESSIONAL EXPERIENCE

AI/ML Research Engineer — Intern

Jan 2026 – Aug 2026

NASA Johnson Space Center - Advanced Operations Concepts Lab

Houston, TX

- Architecting **NASA-AIDA**, a multi-agent, model-agnostic RAG answer engine in direct support of **Artemis Program** crewed mission operations: designed and implemented a **hybrid Neo4j knowledge graph + Qdrant vector database** retrieval stack that replaced dense-only baselines, enabling relationship-aware semantic search across spacecraft flight software, engineering reports, design documents, and Orion telemetry data
- Leading implementation of a **graph ingestion pipeline** for multimodal technical documents into Qdrant/Neo4j, building structured engineering ontologies that map cross-document relationships and enable precision retrieval unavailable to vector-only systems, while driving integration of **graph-based chat history** for persistent contextual reasoning
- Engineering a **hybrid evaluation pipeline** combining **DeepEval** model-based **LLM-as-judge** scoring with statistical NLP metrics across domain-specific QA pairs auto-generated from system documents, directly benchmarking retrieval fidelity and generative quality for mission-operations use cases
- Building an internal **Cursor-like AI coding assistant** with a NASA-secure server that connects the team's LLM and graph/vector DB infrastructure to chunked, embedded, and indexed project codebases, enabling inline diff suggestions and context-aware code completions through local and **ITAR-compliant zero-retention** cloud model endpoints
- Co-authoring a team-led proposal establishing a **NASA-wide AI Validation Lab** — a formal framework for testing and certifying AI applications across the agency by trust level and use case — while overhauling AI certification procedural requirements
- Running NASA-AIDA on-premise on **Nvidia Blackwell/Ampere** hardware, with the local architecture informing **Earth-Independent Operations** research across NASA centers toward future onboard AI integration for extended interplanetary missions

Data/Machine Learning Infrastructure Engineer
Vanderbilt University Medical Center - Data Platform Services

Jan 2022 – Jan 2026
Nashville, TN

- **Spearheaded development of Vanderbilt's first AI Assistant** prototype, architecting enterprise-grade RAG system using Mistral-7B LLM with LangChain on **vector-indexed institutional data** and a production-ready Streamlit inference app serving researchers and clinical teams
- Architected and managed **\$600K annual** Azure cloud infrastructure supporting 100+ Databricks workspaces across Vanderbilt research and corporate departments, serving **400+ researchers, engineers, and data scientists**
- Engineered petabyte-scale data lake using Apache Spark, Azure Data Factory, and Delta Lake with large-scale ETL pipelines processing **150+ data sources** from SQL Server, RESTful APIs, and other systems
- Developed and maintained custom Python pipeline orchestration package with optimized wheel builds, enabling efficient deployment and management of data workflows across **multiple engineering teams**
- Built production DataOps and MLOps infrastructure with Docker containers, GitLab CI/CD pipelines, and Terraform

Theoretical Physicist & Graduate Researcher
Johns Hopkins University

Jan 2019 – May 2020
Baltimore, MD

- Researched the **black hole information paradox** under Professor David Kaplan, developing Python/Mathematica methods to solve coupled nonlinear PDEs from quantum-gravity models governing information flow across event horizons

CLASS Telescope Engineer
Johns Hopkins University

Sept 2017 – Jan 2019
Baltimore, MD

- Designed cryogenic systems for a cosmology telescope (BlueFors dilution refrigerator, Python, SolidWorks) reaching target **temperature of 100mK (-273°C)**, and deployed telescope hardware at a **17,000ft** research site in the Atacama Desert, Chile

RESEARCH & PROJECTS

Catastrophic Forgetting in Mathematical Reasoning

Published Research - arXiv:2512.13706 2025

- Independently authored “Mitigating Catastrophic Forgetting in Mathematical Reasoning Finetuning through Mixed Training”
- Investigated catastrophic forgetting when finetuning FLAN-T5-Base (250M parameters) on DeepMind Mathematics dataset, demonstrating **64.5 percentage point drop** in NLI performance during math-only specialized training
- Developed **mixed training strategies** that completely eliminate catastrophic forgetting while maintaining equivalent math improvement: balanced 1:1 mixing matches math-only **+8.9 point increase** while preserving 86.2% NLI accuracy
- Systematically explored math:NLI mixing ratios from 1:1 to 15:1, demonstrating that even minimal NLI exposure (6.2%) provides effective regularization against catastrophic forgetting; **findings applied to NASA-AIDA continual learning design**

CyberSolve-LinAlg - Finetuned FLAN-T5-Large Model

Published on Hugging Face 2024-2025

- Parallelized multiple Nvidia A100 GPUs to finetune FLAN-T5-Large (780M parameters) on Google DeepMind Mathematics dataset, achieving **90.8% accuracy** on linear equation solving benchmarks
- Implemented distributed training workflow on Azure Databricks processing **2M+** mathematical problems using PyTorch and Hugging Face Accelerate
- Optimized training pipeline with Nvidia Apex fused kernels and AdamW, reducing GPU memory overhead; **created novel evaluation dataset** measuring partial correctness in mathematical reasoning for fine-grained performance analysis beyond binary accuracy
- Deployed interactive app on Hugging Face Spaces using Streamlit with GPU-optimized inference, enabling real-time mathematical problem solving with user-friendly interface

AWARDS & RECOGNITION

NASA Johnson Space Center - Excellence Recognition Award

2026

For exceptional achievement in developing and deploying the first mission critical agentic AI system for Artemis II

Marathon Running

700+ miles in 2025

Top 15% Derby Festival miniMarathon · Top 32% San Diego Marathon

Training into 2026

Johns Hopkins Bloomberg Distinguished Professor STAR Award

2018

\$4000 research grant, nominated by Professor Charles L. Bennett